

ROBOTICS FINAL EXAM PORTFOLIO

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ITAI-2374

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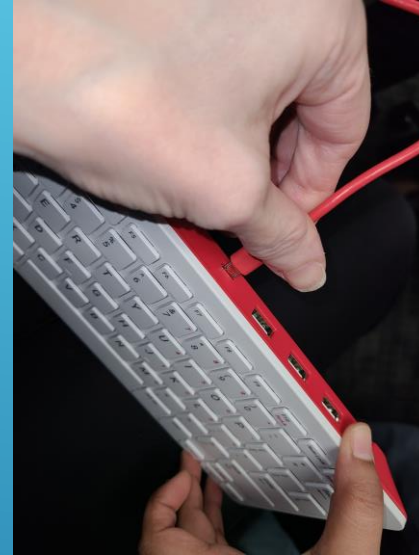
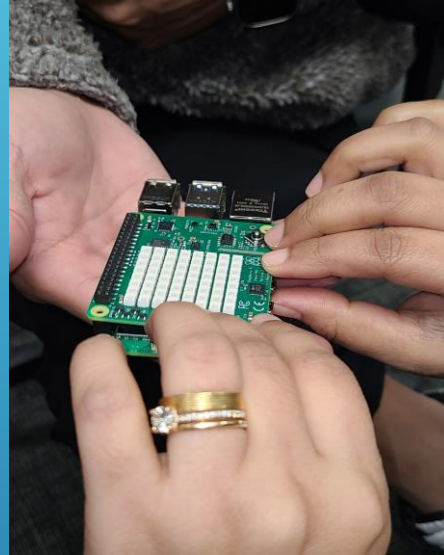
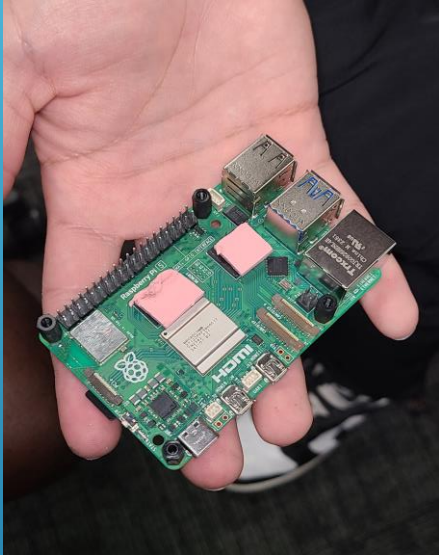
- ▶ Hands-on, interactive learning environment
- ▶ Focus on robotics, AI tools, and embedded systems
- ▶ Emphasis on experimentation, troubleshooting, and applied learning
- ▶ Personal goal: gain exposure to new technologies and practical skills

COURSE OVERVIEW

- ▶ Learned best through hands-on engagement
- ▶ Reading alone is not enough for retention
- ▶ Trial-and-error and physical interaction reinforce understanding
- ▶ This course aligned well with that learning style

LEARNING PHILOSOPHY

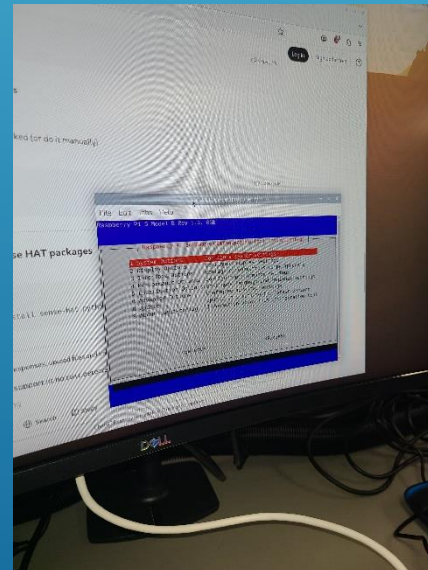
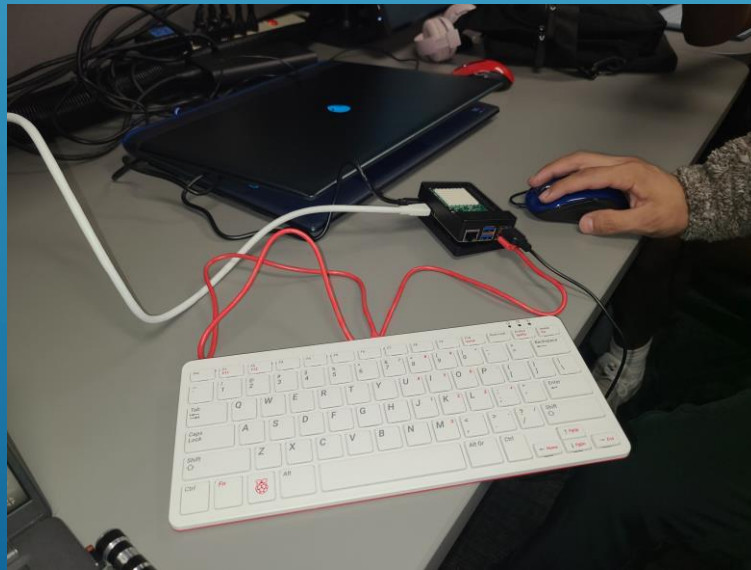
INTRODUCTION TO RASPBERRY PI



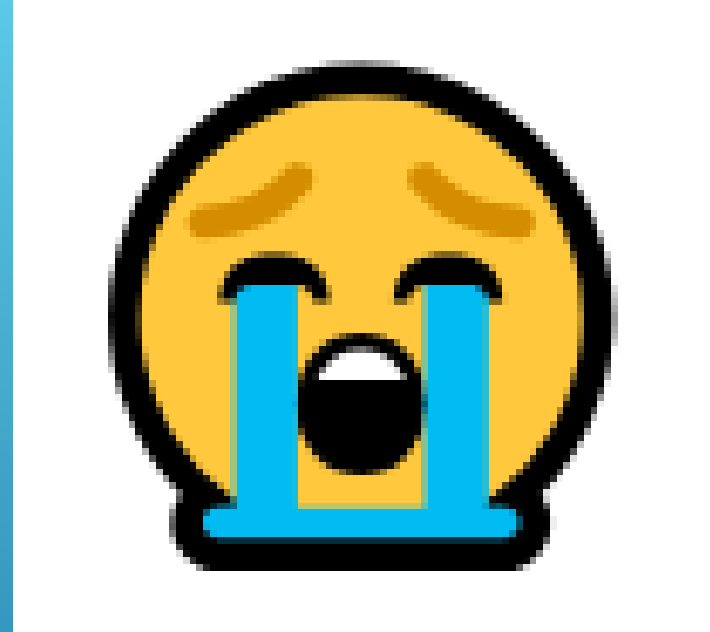
- ▶ Prior familiarity: heard about it but never used one
- ▶ Manual provided foundational knowledge
- ▶ Schematics and component explanations were helpful
- ▶ Frustrating without the device in hand

READING VS. DOING

- ▶ Progressed through ~7 chapters
- ▶ Programming concepts didn't "stick" without practice
- ▶ Realized hands-on work was essential
- ▶ Decided to pause reading and wait for lab time



- ▶ Holiday cancellation
- ▶ Weather-related cancellation
- ▶ Lost 2 out of 16 sessions
- ▶ Increased anticipation for hands-on work



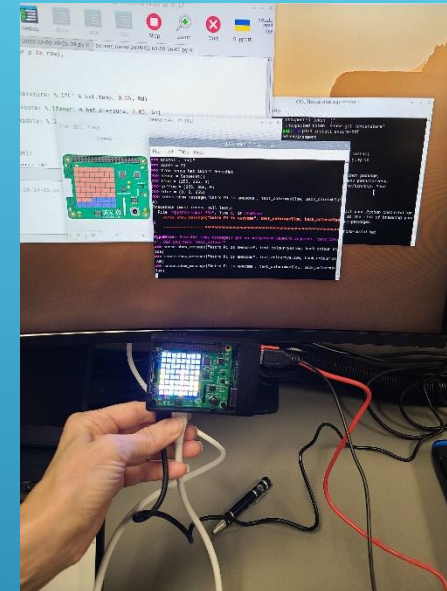
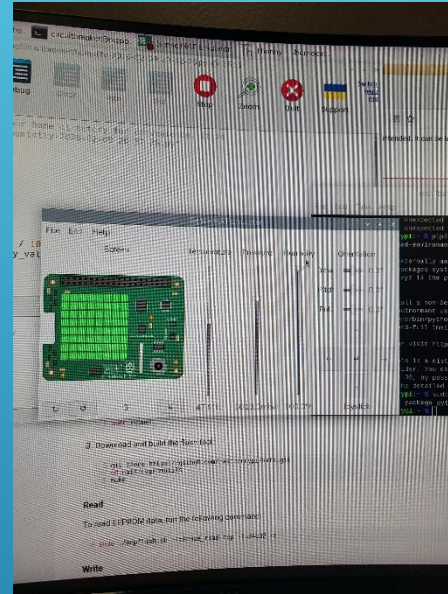
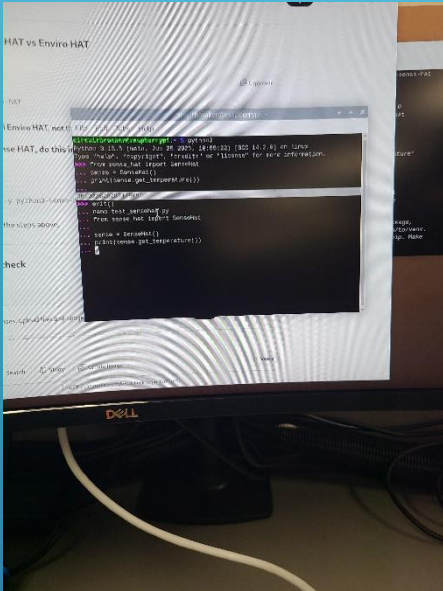
DELAYS AND MISSED CLASSES

RASPBERRY PI LAB EXPERIENCE

- ▶ Assembly challenges: orientation, cables, components
- ▶ Software installation and setup
- ▶ Experimented with LED display code
- ▶ Adjusted scroll speed, colors, brightness
- ▶ Discovered text was scrolling too fast to see



REFLECTION ON RASPBERRY PI



- ▶ Highly engaging and enjoyable
- ▶ Not enough time to explore fully
- ▶ Reinforced preference for hands-on learning
- ▶ Sparked curiosity for future Raspberry Pi projects

- ▶ Initial challenge: registration issues on synthesia.io
- ▶ Multiple emails with tech support
- ▶ Eventually gained access
- ▶ Created short video chatbots
- ▶ Rendering time was longer than expected



**Try out our free AI
Video Tool**

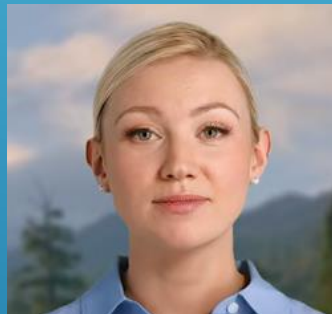
CHATBOT DEVELOPMENT

**Try Synthesia with a
free video.**

Simply type in text and get a free video with an AI avatar in a few clicks. No signup or credit card required.

Create free AI Video →

- ▶ Interesting introduction to conversational AI
- ▶ Limited by free-tier constraints
- ▶ Learned basics of AI-generated multimedia
- ▶ Gained appreciation for backend processing time



REFLECTION ON CHATBOT MODULE

NVIDIA CONFERENCE OVERVIEW

- ▶ Mix of high-level and highly technical content
- ▶ Required pausing and researching unfamiliar terms
- ▶ Expanded understanding of AI and graphics technologies

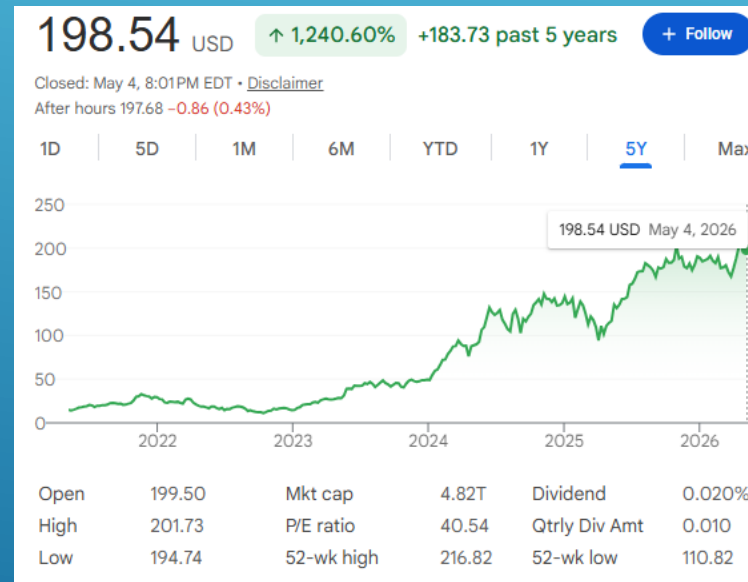


THE “STARS” MOMENT

- ▶ Confusion about “stars” during Jensen Huang’s talk
- ▶ Learned they were GitHub stars (community validation metric)
- ▶ Understood significance of rapid adoption
- ▶ Memorable and humorous learning moment



- ▶ NVIDIA's impact on graphics and AI
- ▶ Exposure to real-world innovation
- ▶ Some sessions beyond current knowledge, but still valuable
- ▶ Reinforced motivation to continue learning
- ▶ Do NOT SELL my NVIDIA stock!



KEY TAKEAWAYS

REAL-WORLD AI APPLICATIONS

- ▶ Language therapy tools using AI
- ▶ Automated tracking of patient progress
- ▶ Insight into practical, human-centered AI
- ▶ Disappointment: replaced a robotics lab session



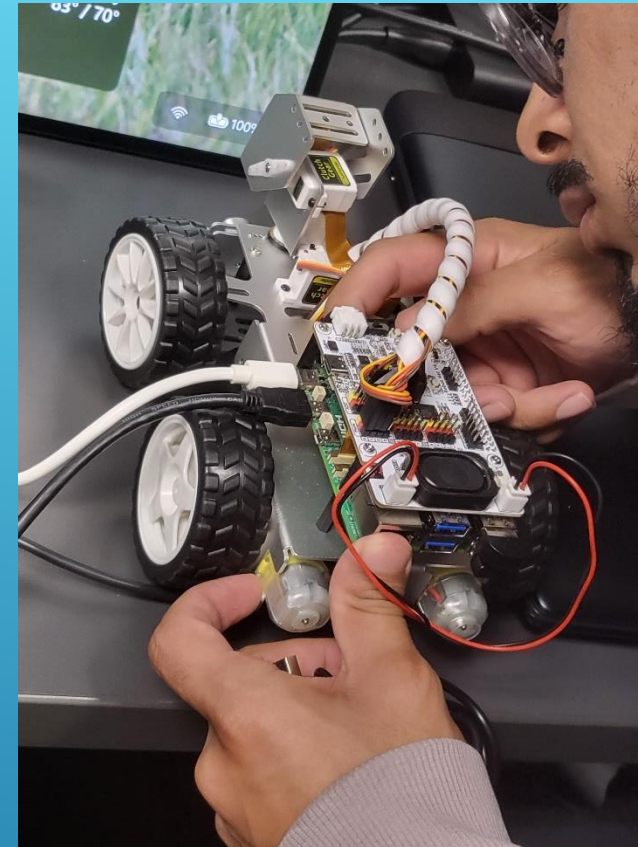
INTEL SPEAKER: JASON DUNN-POTTER

- ▶ Engaging, dynamic presenter
- ▶ Encouraged creative thinking
- ▶ Highlighted innovation happening at all levels
- ▶ Humbling to see peers with advanced side projects
- ▶ Reinforced personal goal: exposure to new ideas
- ▶ Tasty chicken nuggets



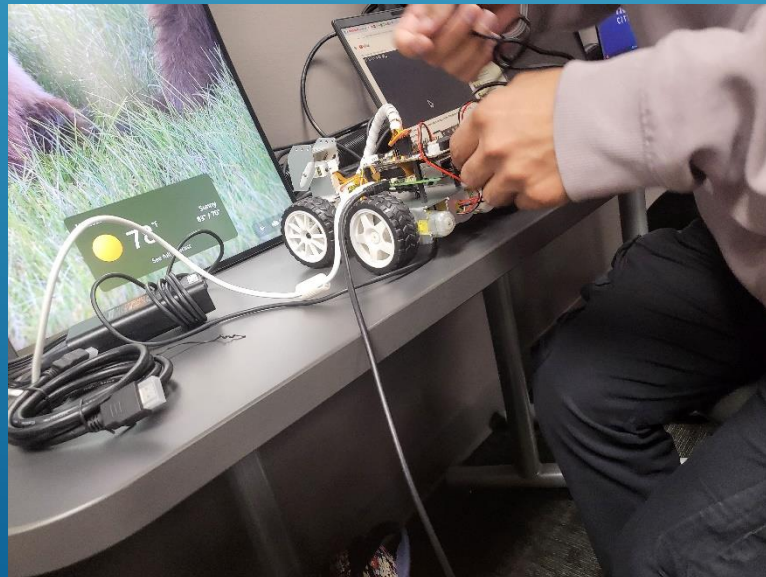
PICAR-X CHALLENGES

- ▶ Missing cables
- ▶ Non-responsive keyboard
- ▶ Weak battery
- ▶ Limited ability to complete tasks
- ▶ Picar-X Has Low Voltage Issues: <https://youtube.com/shorts/OoZG5yq7uzc>
- ▶ Picar-X Moves!: https://youtube.com/shorts/U0XAoPN_09Y?feature=share



REFLECTION ON PICAR-X

- ▶ High expectations, but technical issues limited progress
- ▶ Learned importance of hardware readiness
- ▶ Reinforced troubleshooting skills
- ▶ Experience was disappointing but still educational

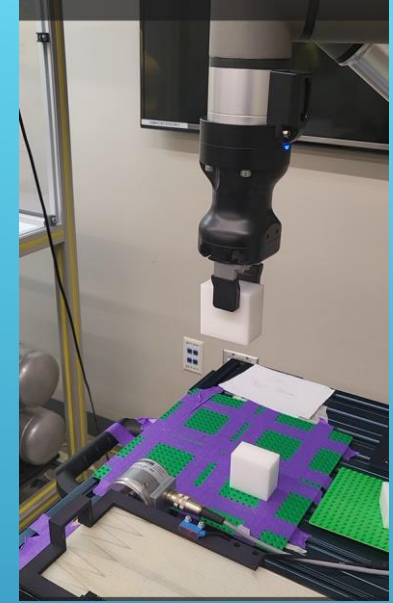


- ▶ Excited to work with a collaborative robot
- ▶ Initial struggle with touchpad controls
- ▶ Learned X, Y, Z coordinate system
- ▶ Watched instructional videos multiple times
- ▶ **Visual:** Cobot arm, coordinate system diagram, screenshots

INTRODUCTION TO THE COBOT

COBOT PROGRESS & REFLECTION

- ▶ Began understanding endpoint-to-endpoint movement
- ▶ Collaborated effectively with partner (Rida)
- ▶ Gained confidence in planning movements
- ▶ Third session cancelled — disappointing
- ▶ Still the MOST rewarding module!!!!
- ▶ Cobot Assembly Line:
<https://youtube.com/shorts/vQ6ZF2lbLPk?feature=share>
- ▶ Trial and Error: <https://youtube.com/shorts/3B6EW21GbeE?feature=share>



- ▶ Course was highly engaging despite cancellations
- ▶ Hands-on learning proved essential for retention
- ▶ Gained exposure to robotics, AI, and embedded systems
- ▶ Inspired to continue exploring technology
- ▶ Appreciative of the practical experience gained
- ▶ Wanted more hands on lab time

FINAL REFLECTION

